



Databases and its Applications

Advanced NoSQL and Emerging Database trends

Pooja T S

Computer Applications

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NewSQL, Serverless DBs, Auto-scaling Storage

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Limitations of Traditional SQL and NoSQL

► Traditional SQL Databases

Strong ACID guarantees and relational model

Vertical scaling (scale-up) is the primary approach

Limited scalability for modern high-traffic applications

► NoSQL Databases

Designed for horizontal scaling and high availability

Often compromise consistency for performance

Not ideal for complex transactions and joins

► Need for a new approach

Combine scalability of NoSQL with consistency of SQL



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What is NewSQL?

► Definition

NewSQL databases preserve the relational model and SQL interface

Designed to scale horizontally like NoSQL systems

► Key idea

ACID compliance without sacrificing scalability

Distributed architecture under the hood

► Why NewSQL matters

Supports modern cloud-native and high-throughput applications



Characteristics of NewSQL Databases

► Core characteristics

- SQL as the primary query language
- Full ACID transaction support
- Distributed, shared-nothing architecture
- High concurrency and low latency

► Architectural techniques

- In-memory processing
- Distributed consensus protocols
- Automatic data partitioning



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Use Cases for NewSQL

► Where NewSQL is suitable

Financial systems requiring strict consistency

E-commerce platforms with high transaction volume

Real-time analytics with transactional guarantees

► Comparison summary

SQL: strong consistency, limited scalability

NoSQL: high scalability, relaxed consistency

NewSQL: scalability + strong consistency



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Serverless Databases – Concept

► What is a Serverless Database?

Database service where users do not manage servers

Infrastructure provisioning handled automatically by the provider

► Developer perspective

Focus only on data and queries

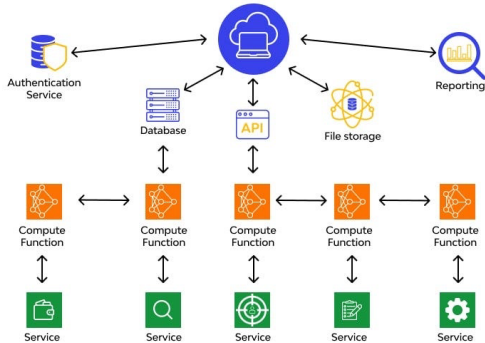
No capacity planning or manual scaling



Serverless Databases

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Serverless Architecture





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Features of Serverless Databases

► Key features

- Automatic scaling based on workload
- Pay-per-use pricing model
- Built-in high availability and fault tolerance

► Advantages

- Reduced operational overhead
- Cost-efficient for variable workloads



Serverless Databases – Execution Model

▶ **How queries are handled**

Queries trigger compute resources on demand

Resources are released after execution

▶ **Workload behaviour**

Supports bursty and unpredictable workloads

No idle resource cost during low usage periods

▶ **Billing principle**

Charges based on actual usage rather than provisioned capacity



► Advantages

- Eliminates database administration overhead
- Automatically adapts to workload changes
- Suitable for microservices and event-driven systems

► Challenges

- Cold-start latency during sudden workload spikes
- Limited low-level control over infrastructure
- Performance tuning options may be restricted



► What is Auto-scaling Storage?

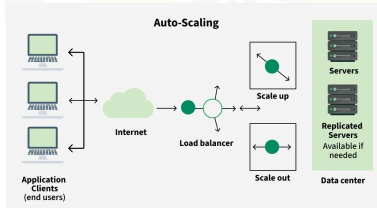
Storage capacity automatically increases or decreases

No manual disk or volume management

► Why auto-scaling is important

Data growth is unpredictable

Prevents over-provisioning and storage waste





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Auto-scaling Storage – How It Works

► **Monitoring mechanisms**

Continuous tracking of data growth and access patterns
Threshold-based or predictive scaling decisions

► **Storage behaviour**

Expands transparently without application changes
Ensures uninterrupted read/write operations



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Auto-scaling Storage – Benefits

► **Operational benefits**

- Removes the need for capacity planning
- Prevents storage-related failures

► **Cost benefits**

- Pay only for storage actually used
- Avoids over-provisioning and wastage

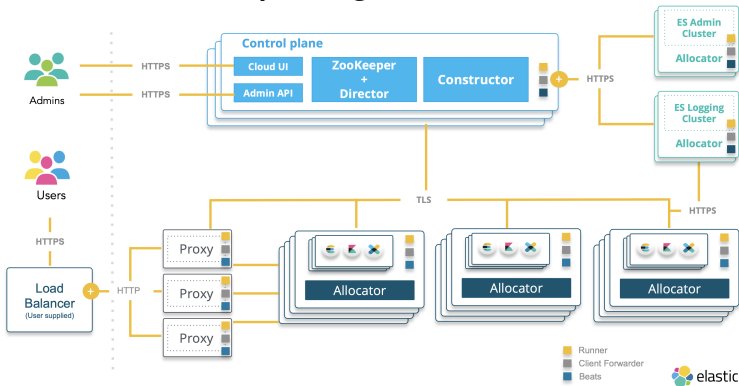
► **Scalability**

- Supports applications with rapid and unpredictable data growth



Auto-scaling Storage

Elastic Cloud Enterprise high level architecture





► How they work together

Serverless databases rely on auto-scaling storage

Compute and storage scale independently

► Impact on modern systems

Enables elastic, cloud-native applications

Ideal for microservices and event-driven architectures



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CELEBRATING 50 YEARS

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